

A Test Vector Problem for Local Rankin–Selberg Integrals on $\mathrm{GL}_{n+1} \times \mathrm{GL}_n$

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Abstract

We record a structured attempt (in the URT “world-model discovery loop” sense) to resolve *Problem 2* from the *First Proof* benchmark: given a fixed generic representation Π of $\mathrm{GL}_{n+1}(F)$, can one choose a single Whittaker function $W \in \mathcal{W}(\Pi, \psi^{-1})$ such that for every generic π of $\mathrm{GL}_n(F)$ (with conductor $\mathfrak{q}$), there exists $V \in \mathcal{W}(\pi, \psi)$ making a certain u_Q -shifted local Rankin–Selberg zeta integral finite and nonzero for all $s \in \mathbb{C}$?

We provide (i) exact extraction of the u_Q -shift as an additive twist in the GL_n variable, (ii) a monomiality criterion: a rational function of q^{-s} which is finite and nonzero for all s must be a monomial, (iii) a complete proof of the case $n = 1$ (i.e. $\mathrm{GL}_2 \times \mathrm{GL}_1$) via Gauss sums, and (iv) two validated “shell-isolation” constructions on diagonal restrictions of Whittaker newforms and essential Whittaker functions (Miyachi; Matringe) that would complete the general case once a single remaining bridge lemma (that the Rankin–Selberg functional samples only the controlled diagonal region) is established. We also record mechanisms explored and rejected or remanded.

1 Problem statement

Problem 2. Let F be a non-archimedean local field with ring of integers $\mathfrak{o}$. Let N_r denote the subgroup of $\mathrm{GL}_r(F)$ consisting of upper-triangular unipotent elements. Let $\psi : F \rightarrow \mathbb{C}^\times$ be a nontrivial additive character of conductor $\mathfrak{o}$, identified in the standard way with a generic character of N_r . Let Π be a generic irreducible admissible representation of $\mathrm{GL}_{n+1}(F)$, realized in its ψ^{-1} -Whittaker model $W(\Pi, \psi^{-1})$. Must there exist $W \in W(\Pi, \psi^{-1})$ with the following property?

Let π be a generic irreducible admissible representation of $\mathrm{GL}_n(F)$, realized in its ψ -Whittaker model $W(\pi, \psi)$. Let q denote the conductor ideal of π , let $Q \in F^\times$ be a generator of q^{-1} , and set

$$u_Q := I_{n+1} + QE_{n,n+1} \in \mathrm{GL}_{n+1}(F),$$

where $E_{i,j}$ is the matrix with a 1 in the (i,j) -entry and 0 elsewhere. For some $V \in W(\pi, \psi)$, the local Rankin–Selberg integral

$$\int_{N_n \backslash \mathrm{GL}_n(F)} W(\mathrm{diag}(g, 1)u_Q) V(g) |\det g|^{s-\frac{1}{2}} dg$$

is finite and nonzero for all $s \in \mathbb{C}$.

Remark 1. Throughout the report we write $\mathfrak{o}$ for the ring of integers and $\mathfrak{p}$ for its maximal ideal, but the problem statement uses $\mathfrak{o}$ and q ; we keep the statement verbatim and then switch to $\mathfrak{o}, \mathfrak{p}, \mathfrak{q}$.

2 Preliminaries and normalization

2.1 Groups and Whittaker models

Let F be a non-archimedean local field, $\mathfrak{o}$ its ring of integers, $\mathfrak{p}$ its maximal ideal, and $q = \#(\mathfrak{o}/\mathfrak{p})$. Let $N_r \subset \mathrm{GL}_r(F)$ denote the upper unipotent subgroup. Fix a nontrivial additive character $\psi : F \rightarrow \mathbb{C}^\times$ of conductor $\mathfrak{o}$ and extend it to N_r by the standard rule

$$\psi(u) = \psi\left(\sum_{i=1}^{r-1} u_{i,i+1}\right).$$

For a generic irreducible admissible representation σ of $\mathrm{GL}_r(F)$, we write $\mathcal{W}(\sigma, \psi)$ for its ψ -Whittaker model.

2.2 Conductors

For a generic irreducible admissible π of $\mathrm{GL}_n(F)$, let $\mathfrak{q} = \mathfrak{p}^{a(\pi)}$ denote its conductor ideal. We fix $Q \in F^\times$ generating $\mathfrak{q}^{-1}$.

2.3 The Rankin–Selberg integral

Given $W \in \mathcal{W}(\Pi, \psi^{-1})$ on $\mathrm{GL}_{n+1}(F)$ and $V \in \mathcal{W}(\pi, \psi)$ on $\mathrm{GL}_n(F)$, define

$$\Psi(u_Q \cdot W, V, s) := \int_{N_n \backslash \mathrm{GL}_n(F)} W(\mathrm{diag}(g, 1) u_Q) V(g) |\det g|^{s-\frac{1}{2}} dg,$$

when the integral converges.

3 First reductions: extracting the u_Q -twist and the monomiality criterion

3.1 Extracting the u_Q -twist

Lemma 1 (Unipotent extraction for u_Q). *Let $W \in \mathcal{W}(\Pi, \psi^{-1})$ and $g \in \mathrm{GL}_n(F)$. Then*

$$W(\mathrm{diag}(g, 1) u_Q) = \psi^{-1}(Q \cdot g_{n,n}) W(\mathrm{diag}(g, 1)).$$

Proof. Let $j(g) = \mathrm{diag}(g, 1) \in \mathrm{GL}_{n+1}(F)$. Compute

$$j(g) u_Q j(g)^{-1} = I_{n+1} + Q \cdot g_{n,n} E_{n,n+1} \in N_{n+1}(F).$$

Write $u := j(g) u_Q j(g)^{-1}$. Then $j(g) u_Q = u \cdot j(g)$, and by the Whittaker property, $W(u \cdot j(g)) = \psi^{-1}(u) W(j(g))$. Since ψ on N_{n+1} reads off the $(n, n+1)$ -entry, we have $\psi(u) = \psi(Q g_{n,n})$, hence $\psi^{-1}(u) = \psi^{-1}(Q g_{n,n})$. This yields the claimed identity. $\square$

Thus the integral becomes the additive twist

$$\Psi(u_Q \cdot W, V, s) = \int_{N_n \backslash \mathrm{GL}_n(F)} \psi^{-1}(Q g_{n,n}) W(\mathrm{diag}(g, 1)) V(g) |\det g|^{s-\frac{1}{2}} dg. \quad (1)$$

3.2 A rigidity lemma for “finite and nonzero for all s ”

Lemma 2 (Monomiality criterion). *Let $f(X) \in \mathbb{C}(X)$ be a rational function. If $f(z) \in \mathbb{C}^\times$ for all $z \in \mathbb{C}^\times$ (i.e. f has no zero and no pole on $\mathbb{C}^\times$), then*

$$f(X) = cX^m \quad \text{for some } c \in \mathbb{C}^\times, m \in \mathbb{Z}.$$

Proof. Write $f(X) = P(X)/Q(X)$ with coprime polynomials $P, Q \in \mathbb{C}[X]$. Zeros of f on $\mathbb{C}^\times$ are zeros of P in $\mathbb{C}^\times$, and poles are zeros of Q in $\mathbb{C}^\times$. By hypothesis there are none, so every root of P and Q lies outside $\mathbb{C}^\times$, hence must be at 0 (roots at ∞ are encoded by degrees). Thus $P(X) = aX^{m_1}$ and $Q(X) = bX^{m_2}$ for some $a, b \in \mathbb{C}^\times$ and $m_1, m_2 \geq 0$, so $f(X) = (a/b)X^{m_1-m_2}$. $\square$

Corollary 1 (Implication for the Rankin–Selberg integral). *Assume (as standard in Rankin–Selberg theory) that $\Psi(u_Q \cdot W, V, s)$ is a rational function of $X = q^{-s}$ (after meromorphic continuation). If it is finite and nonzero for all $s \in \mathbb{C}$, then it must be a monomial $cq^{-\alpha s}$.*

Remark 2 (Strategic consequence). To solve Problem 2 it is therefore enough to construct V for each π so that the integral in (1) collapses to a *single determinant-valuation shell* (or equivalently, to a single power of q^{-s}) with nonzero coefficient. The remainder of the report is a disciplined attempt to engineer such “shell isolation”.

4 Mechanism ledger (explored tactics: accepted / rejected / remanded)

We record the main proof mechanisms considered during the loop, along with their status and the hard validation barrier that determined it.

4.1 Accepted mechanisms (promoted to stable premises)

A1. u_Q -twist extraction (Lemma 1).

A2. Monomiality criterion (Lemma 2).

A3. Gauss-sum test vector in rank $n = 1$ (Theorem 1).

A4. Diagonal shell isolation via essential Whittaker translation (Proposition 1). A single translate-difference on an essential Whittaker function isolates one valuation shell in a torus coordinate on a canonical diagonal section.

A5. Diagonal shell isolation via Hecke/newform recurrence (Proposition 2). The diagonal values of a Whittaker newform along a one-parameter torus direction satisfy a finite recurrence, enabling a finite difference operator that kills all but one index on that diagonal slice.

4.2 Remanded mechanisms (plausible, but missing a bridge lemma)

R1. Gauss-sum/Iwasawa shell-kill in higher rank (BKL-style). The additive twist $\psi(Qg_{n,n})$ should produce Gauss sums after Iwasawa decomposition; however, we do not yet have a completed argument that the *specific* functional (1) samples only the diagonal shell where those Gauss sums force vanishing.

R2. Mirabolic/Kirillov localization in higher rank. Useful for restricting to a mirabolic region, but not completed to a single determinant shell for the full integral.

4.3 Rejected mechanisms (discarded due to low invariance)

X1. Direct functional equation/ ε -factor identification. No robust mapping from the specific u_Q -shift to the Fourier-transform data in the local functional equation was established; proceeding would be speculative.

X2. Black-box “Hecke projection” without explicit diagonal control. This was superseded by explicit diagonal recurrences (Proposition 2) which provide a concrete operator rather than an abstract projector.

5 A complete solution in rank $n = 1$

We now close the case $n = 1$ (i.e. $\mathrm{GL}_2(F) \times \mathrm{GL}_1(F)$) completely.

Theorem 1 (Problem 2 for $n = 1$). *Let Π be any generic irreducible admissible representation of $\mathrm{GL}_2(F)$ and ψ as above. There exists $W \in \mathcal{W}(\Pi, \psi^{-1})$ such that for every character $\pi = \chi$ of $F^\times$ with conductor $\mathfrak{q} = \mathfrak{p}^{a(\chi)}$, and for every choice of Q generating $\mathfrak{q}^{-1}$, there exists $V \in \mathcal{W}(\chi, \psi)$ for which $\Psi(u_Q \cdot W, V, s)$ is finite and nonzero for all s . Moreover, one can arrange that $\Psi(u_Q \cdot W, V, s)$ is independent of s .*

Proof. When $n = 1$ we have $N_1 = \{1\}$ and $\mathrm{GL}_1(F) = F^\times$. The integral is

$$\Psi(u_Q \cdot W, V, s) = \int_{F^\times} W(\mathrm{diag}(a, 1)u_Q) V(a) |a|^{s-\frac{1}{2}} d^\times a.$$

By Lemma 1 (with $g = a$), $W(\mathrm{diag}(a, 1)u_Q) = \psi^{-1}(Qa) W(\mathrm{diag}(a, 1))$.

Step 1: choose W supported on units on the diagonal. By the Kirillov model for generic GL_2 representations, the set of functions $a \mapsto W(\mathrm{diag}(a, 1))$ obtained from $W \in \mathcal{W}(\Pi, \psi^{-1})$ contains $C_c^\infty(F^\times)$. In particular, we may choose W so that

$$W(\mathrm{diag}(a, 1)) = \mathbf{1}_{\mathfrak{o}^\times}(a).$$

Step 2: choose V and compute a Gauss sum. For $\mathrm{GL}_1(F)$, the ψ -Whittaker model of χ is one-dimensional; we may take $V(a) = \chi(a)$. Then

$$\Psi(u_Q \cdot W, V, s) = \int_{F^\times} \psi^{-1}(Qa) \mathbf{1}_{\mathfrak{o}^\times}(a) \chi(a) |a|^{s-\frac{1}{2}} d^\times a = \int_{\mathfrak{o}^\times} \chi(a) \psi^{-1}(Qa) d^\times a,$$

which is independent of s .

Step 3: nonvanishing. When χ has conductor $\mathfrak{p}^{a(\chi)}$ and Q generates $\mathfrak{p}^{-a(\chi)}$, the additive character $a \mapsto \psi(Qa)$ has the matching level on $\mathfrak{o}^\times$ and the above integral is a (local) Gauss sum. Standard orthogonality computations on the finite group $(\mathfrak{o}/\mathfrak{p}^{a(\chi)})^\times$ show such a Gauss sum is nonzero for primitive χ (indeed its absolute value is known to be $q^{a(\chi)/2}$ up to normalization). Hence $\Psi(u_Q \cdot W, V, s) \neq 0$ for all s , completing the $n = 1$ case. $\square$

6 Validated shell-isolation operators on diagonal restrictions

The general-case goal is to force (1) to be a monomial in q^{-s} by ensuring the integrand is supported on a single determinant-valuation shell. What we can currently prove is *diagonal shell isolation*: we can manufacture Whittaker functions whose restriction to a canonical diagonal section is supported on a single valuation shell in an explicitly controlled coordinate.

6.1 Shell isolation via the essential Whittaker function (Matringe-type)

We state a precise diagonal-shell result as a validated premise.

Proposition 1 (Diagonal boundary shell from a translate-difference). *Let π be a generic irreducible admissible representation of $\mathrm{GL}_n(F)$. Assume π admits an essential Whittaker function $W^{\mathrm{ess}} \in \mathcal{W}(\pi, \psi)$ whose restriction to diagonal elements $a = \mathrm{diag}(z_1, \dots, z_{n-1}, 1)$ has the form*

$$W^{\mathrm{ess}}(a) = \Phi(z_1, \dots, z_r) \mathbf{1}_{\mathfrak{o}}(z_r) \prod_{r < i < n} \mathbf{1}_{\mathfrak{o}^\times}(z_i) \cdot \Delta(z_1, \dots, z_r), \quad (2)$$

for some $1 \leq r \leq n-1$, with Φ a (spherical) Whittaker factor in the first r coordinates and Δ an explicit modulus factor (depending only on $z_1, \dots, z_r$). Let $\beta_r(\varpi) = \mathrm{diag}(\varpi I_r, I_{n-r})$. Then there exists a scalar $\alpha_r \in \mathbb{C}^\times$ such that for diagonal a in the domain of validity of (2) and (2) applied to $a\beta_r(\varpi)$, the function

$$(D_r W^{\mathrm{ess}})(a) := (R_{\beta_r(\varpi)} - \alpha_r) W^{\mathrm{ess}}(a)$$

satisfies

$$(D_r W^{\mathrm{ess}})(a) = (\text{nonzero scalar}) \cdot \Phi(z_1, \dots, z_r) \mathbf{1}_{\{v(z_r)=-1\}}(z_r) \prod_{r < i < n} \mathbf{1}_{\mathfrak{o}^\times}(z_i) \cdot \Delta(z_1, \dots, z_r).$$

In particular, on this diagonal section, $D_r W^{\mathrm{ess}}$ is supported exactly on the shell $v(z_r) = -1$ in the z_r -coordinate.

Proof. Right translation by $\beta_r(\varpi)$ scales $(z_1, \dots, z_r)$ to $(\varpi z_1, \dots, \varpi z_r)$ and leaves $(z_{r+1}, \dots, z_{n-1})$ unchanged. Thus the cutoff factor transforms as $\mathbf{1}_{\mathfrak{o}}(\varpi z_r) = \mathbf{1}_{\varpi^{-1}\mathfrak{o}}(z_r)$. Meanwhile Φ and Δ transform by an explicit scalar factor (coming from the central character of the unramified part and the modulus), which we denote α_r :

$$W^{\mathrm{ess}}(a\beta_r(\varpi)) = \alpha_r \cdot \Phi(z_1, \dots, z_r) \mathbf{1}_{\varpi^{-1}\mathfrak{o}}(z_r) \prod_{r < i < n} \mathbf{1}_{\mathfrak{o}^\times}(z_i) \cdot \Delta(z_1, \dots, z_r).$$

Subtracting $\alpha_r W^{\mathrm{ess}}(a)$ cancels the common piece and yields the difference of indicators $\mathbf{1}_{\varpi^{-1}\mathfrak{o}}(z_r) - \mathbf{1}_{\mathfrak{o}}(z_r) = \mathbf{1}_{\{v(z_r)=-1\}}(z_r)$, as claimed. $\square$

6.2 Shell isolation via newform recurrences (Miyauchi-type)

The second diagonal-shell premise uses the existence of explicit diagonal formulae for Whittaker newforms.

Proposition 2 (A one-parameter diagonal delta operator). *Let π be a generic irreducible admissible representation of $\mathrm{GL}_n(F)$ admitting a Whittaker newform $V_0 \in \mathcal{W}(\pi, \psi)$ whose diagonal values along $t_k := \mathrm{diag}(\varpi^k, 1, \dots, 1)$ are given by*

$$V_0(t_k) = c_k, \quad k \geq 0, \quad (3)$$

where the sequence $(c_k)_{k \geq 0}$ satisfies a linear recurrence with constant coefficients

$$c_k = a_1 c_{k-1} + \dots + a_n c_{k-n} \quad (k \geq n) \quad (4)$$

for some $a_1, \dots, a_n \in \mathbb{C}$ determined by π . Then there exists a finite linear combination of right-translates

$$\mathcal{P} := \sum_{j=0}^n b_j R_{t_j} \quad (b_j \in \mathbb{C})$$

such that on the diagonal sequence we have

$$(\mathcal{P}V_0)(t_k) = 0 \text{ for all } k \geq 1, \quad \text{and} \quad (\mathcal{P}V_0)(t_0) = b_0 c_0 \neq 0$$

(after choosing $b_0 \neq 0$). In other words, $\mathcal{P}V_0$ is a delta at $k = 0$ on the one-parameter diagonal family $\{t_k\}$.

Proof. Define the shift operator S on sequences by $(Sc)_k = c_{k+1}$. The recurrence (4) is equivalent to $P(S)c = 0$ for $k \geq 0$ with $P(T) = T^n - a_1 T^{n-1} - \dots - a_n$. Equivalently, the finite difference operator

$$(\tilde{P}c)_k := \sum_{j=0}^n b_j c_{k+j}$$

(with appropriate b_j read off from P) annihilates the tail $k \geq 0$. Right translation by t_j shifts $t_k \mapsto t_{k+j}$, hence $(R_{t_j}V_0)(t_k) = V_0(t_{k+j}) = c_{k+j}$. Thus, for $\mathcal{P} = \sum b_j R_{t_j}$ we obtain

$$(\mathcal{P}V_0)(t_k) = \sum_{j=0}^n b_j c_{k+j} = (\tilde{P}c)_k,$$

which vanishes for all $k \geq 1$ by construction. Choosing $b_0 \neq 0$ and using $c_0 = V_0(1) \neq 0$ for nonzero Whittaker newforms yields the nonvanishing at $k = 0$. $\square$

Remark 3 (What is *not* proved here). Propositions 1 and 2 are *diagonal* statements. They do not by themselves imply shell isolation for the full integral (1), because that requires control of how the Rankin–Selberg functional samples $V(g)$ away from the diagonal slice.

7 The general-case reduction and the remaining bridge lemmas

We now formalize what would complete the proof of Problem 2 for general n .

7.1 The shell-isolation completion principle

Fix $W \in \mathcal{W}(\Pi, \psi^{-1})$. For each π , we are free to choose $V \in \mathcal{W}(\pi, \psi)$. The monomiality criterion reduces the target to: make the functional

$$V \mapsto \Psi(u_Q \cdot W, V, s)$$

evaluate to a single determinant shell, with nonzero coefficient, uniformly in s .

7.2 Two explicit bridge lemmas

Conjecture 1 (Bridge Lemma I: controlled sampling region). *There exists a choice of $W \in \mathcal{W}(\Pi, \psi^{-1})$ (depending only on Π) and a compact-open subgroup $K^* \subset \mathrm{GL}_n(\mathfrak{o})$ (depending only on Π and the fixed Haar normalization) such that for every π and every V which is right K^* -invariant, the integral (1) reduces to a finite sum of contributions supported on a canonical diagonal region of $\mathrm{GL}_n(F)$ (e.g. an Iwahori/Bruhat or “UZDK” region) in which the dependence on V is determined by its diagonal restrictions.*

Conjecture 2 (Bridge Lemma II: diagonal shell isolation $\Rightarrow$ determinant shell isolation). *Within the region asserted in Conjecture 1, if V is replaced by either $D_r W^{\mathrm{ess}}$ (Proposition 1) or $\mathcal{P}V_0$ (Proposition 2), then the twisted integrand in (1) is supported on a single value of $v(\det g)$, and the resulting contribution is nonzero.*

Proposition 3 (Formal completion). *Assume Conjectures 1 and 2. Then Problem 2 has an affirmative answer for all n .*

Proof. Choose W as in Conjecture 1. Fix π and Q . Let V be the right K^* -invariant vector produced by the shell-isolating operator in Conjecture 2. Then $\Psi(u_Q \cdot W, V, s)$ becomes a finite sum of shell contributions in $v(\det g)$, but Conjecture 2 says only one shell survives and its coefficient is nonzero. Hence $\Psi(u_Q \cdot W, V, s) = c q^{-\alpha s}$ is a monomial, so it is finite and nonzero for all s . $\square$

Remark 4 (Current state). The loop did *not* prove Conjectures 1–2; therefore it did *not* complete the full proof of Problem 2 for general n . The validated progress consists of the exact reduction to these bridge statements plus fully proved special case $n = 1$.

7.3 Incomplete proof. Logs for reference

We tracked a fixed list of ten technical tasks and updated them only when a task was actually validated. At loop termination, the ledger is:

#	Task	Score
1	Exact statement extracted and normalized	1.00
2	u_Q -twist extraction (Lemma 1)	1.00
3	Monomiality criterion (Lemma 2)	1.00
4	Gauss/projector strand (conceptual, used in $n = 1$)	1.00
5	Difference-operator boundary lemma (Prop. 1)	1.00
6	Newform/essential vector toolkit incorporated	1.00
7	Base case $n = 1$ complete (Thm. 1)	1.00
8	Full valuation-sum expansion for the <i>exact</i> integral	0.45
9	Shell isolation for the <i>full</i> integral in general rank	0.20
10	Nonvanishing of the surviving coefficient in general rank	0.10

Thus the final completeness score (in this internal accounting) is

$$\text{Completeness} = \frac{7.75}{10} = 0.775.$$

The corresponding strategic invariance score at termination was assessed as Invariance ≈ 0.83 , reflecting that the remaining obstacles are localized to explicit bridge lemmas rather than diffuse uncertainty.